Sensors

Available data in each dataset

	accelerometer	gyroscope	magnetometer	EMG
HuGaDB				
McGill				
MAREA				
Vajdi				
OU-ISIR ISD 1				
OU-ISIR ISD 2 IMUZ				
OU-ISIR ISD 2 Smartphone				
OU-ISIR SAID				
DBHuGa				

Accelerometer
Gyroscope
Magnetometer

no information given for sensor 3 HuGaDB EMG

Internal Coordinate System

int.co.sys.id	x	y	z	rotation_me	Sensor_ids + Datenbank
0	r	sup	ant	[[1,0,0],[0,1,0],[0,0,1]]	1 - HuGaDB, 2 - HuGaDB, 4 - McGill, 5 - McGill, 8 - ISIR1+2+SAID (IMUZ), 9 - ISIR1+2+SAID (IMUZ), 14 - Marea
1	sup	l	ant	[[0,-1,0],[1,0,0],[0,0,1]]	10 - ISIR2 (Smartphone) 11 - DBHuGa, 12 - DBHuGa, 13 - DBHuGa
2	r	ant	sup	[[1,0,0],[0,0,1],[0,1,0]]	6 - Vajdi, 7 - Vajdi

Sensor Location

SL_ID	Location	Rotation	DB/sensors
1	top of left foot	[[0,1,0],[0,0,-1],[-1,0,0]]	HuGaDB, HuGaDB
2	top of right foot	[[0,1,0],[0,0,-1],[-1,0,0]]	HuGaDB, HuGaDB
3	left Shin	[[0,1,0],[1,0,0],[0,0,-1]]	HuGaDB, HuGaDB
4	right Shin	[[0,1,0],[1,0,0],[0,0,-1]]	HuGaDB, HuGaDB
5	left thigh front	[[0,1,0],[1,0,0],[0,0,-1]]	HuGaDB, HuGaDB
6	right thigh front	[[0,1,0],[1,0,0],[0,0,-1]]	HuGaDB, HuGaDB
7	left vastus lateralis		HuGaDB
8	right vastus lateralis		HuGaDB
9	left front pocket	[[1,0,0],[0,-1,0],[0,0,-1]]	McGill, Vajdi, Vajdi, McGill
10	left back pocket	[[-1,0,0],[0,-1,0],[0,0,1]]	McGill, McGill
11	right front pocket	[[1,0,0],[0,-1,0],[0,0,-1]]	McGill, McGill
12	right waist	[[0,0,1],[0,1,0],[-1,0,0]]	ISIR2+SAID, ISIR2+SAID

13	right waist	[[0,-1,0],[-1,0,0],[0,0,-1]]	Vajdi, Vajdi
14	center waist back	[[-1,0,0],[0,1,0],[0,0,-1]]	ISIR1+2+SAID, ISIR1+2+SAID
15	center waist back	[[1,0,0],[0,-1,0],[0,0,-1]]	ISIR2 (smartphone)
16	left waist	[[0,0,-1],[0,1,0],[1,0,0]]	ISIR2+SAID, ISIR2+SAID
17	left wrist	[[1,0,0],[0,0,1],[0,-1,0]]	DBHuGa, DBHuGa, DBHuGa
18	right wrist	[[0,0,1],[0,-1,0],[1,0,0]]	Marea
19	center waist front	[[-1,0,0],[0,-1,0],[0,0,1]]	Marea
20	left ankle	[[0,0,-1],[0,-1,0],[-1,0,0]]	Marea**| placement here is arbitrary!**
21	right ankle	[[0,0,1],[0,-1,0],[1,0,0]]	Marea
22	left shin	[[1,0,0],[0,1,0],[0,0,1]]	DBHuGa, DBHuGa, DBHuGa
23	right shin	[[1,0,0],[0,1,0],[0,0,1]]	DBHuGa, DBHuGa, DBHuGa
24	left thigh front	[[1,0,0],[0,1,0],[0,0,1]]	DBHuGa, DBHuGa, DBHuGa
25	right thigh front	[[1,0,0],[0,1,0],[0,0,1]]	DBHuGa, DBHuGa, DBHuGa
26	center waist back	[[1,0,0],[0,-1,0],[0,0,-1]]	DBHuGa, DBHuGa, DBHuGa

From here on our different modalities are not listed separately, since the calculation stay the same.

World Coordinate System

w_id = wordl_coordinate_system_id

a_id = alignment_id

w_id	a_id	**x**	**y**	**z**	**both**	**DB&Sensors**
1	0	sup	r	post	[[0,1,0],[1,0,0],[0,0,-1]]	HuGaDB - other acc + gyro, OU-ISIR center back (smartphone), DBHuGA - center back
2	1	post	r	Inf	[[0,1,0],[0,0,-1],[-1,0,0]]	HuGaDB - feet acc + gyro
3	2	r	inf	post	[[1,0,0],[0,-1,0],[0,0,-1]]	McGill - front pockets
4	3	l	inf	ant	[[-1,0,0],[0,-1,0],[0,0,1]]	McGill - back pocket, MAREA - waist front center
5	4	post	l	Inf	[[0,-1,0],[0,0,-1],[-1,0,0]]	Vajdi - right waist
6	5	r	post	Inf	[[1,0,0],[0,0,-1],[0,-1,0]]	Vajdi - left front pocket
7	6	post	sup	R	[[0,0,1],[0,1,0],[-1,0,0]]	OU-ISIR 2+SAID right waist
8	7	l	sup	post	[[-1,0,0],[0,1,0],[0,0,-1]]	OU-ISIR 1+2+SAID center back
9	8	ant	sup	L	[[0,0,-1],[0,1,0],[1,0,0]]	OU-ISIR 2+SAID left waist
10	9	sup	l	ant	[[0,-1,0],[1,0,0],[0,0,1]]	DBHuGA - thigh and shin
11	10	sup	post	L	[[0,0,-1],[1,0,0],[0,-1,0]]	DBHuGA - left wrist
12	11	ant	inf	R	[[0,0,1],[0,-1,0],[1,0,0]]	MAREA - right ankle, right wrist
13	12	post	inf	L	[[0,0,-1],[0,-1,0],[-1,0,0]]	MAREA - left ankle

Combined Internal Coordinate System, Sensor Location and World Coordinate System

x	y	z	int.coord.sys.	loc.rot	world coordinate system	DB&Sensors
sup	r	post	[[1,0,0],[0,1,0],[0,0,1]]	[[0,1,0],[1,0,0],[0,0,-1]]	[[0,1,0],[1,0,0],[0,0,-1]]	HuGaDB - other acc + gyro
post	r	inf	[[1,0,0],[0,1,0],[0,0,1]]	[[0,1,0],[0,0,-1],[-1,0,0]]	[[0,1,0],[0,0,-1],[-1,0,0]]	HuGaDB - feet acc + gyro
r	inf	post	[[1,0,0],[0,1,0],[0,0,1]]	[[1,0,0],[0,-1,0],[0,0,-1]]	[[1,0,0],[0,-1,0],[0,0,-1]]	McGill - front pockets
l	inf	ant	[[1,0,0],[0,1,0],[0,0,1]]	[[-1,0,0],[0,-1,0],[0,0,1]]	[[-1,0,0],[0,-1,0],[0,0,1]]	McGill - back pocket, MAREA - waist front center
ant	sup	l	[[1,0,0],[0,1,0],[0,0,1]]	[[0,0,-1],[0,1,0],[1,0,0]]	[[0,0,-1],[0,1,0],[1,0,0]]	OU-ISIR left waist
l	sup	post	[[1,0,0],[0,1,0],[0,0,1]]	[[-1,0,0],[0,1,0],[0,0,-1]]	[[-1,0,0],[0,1,0],[0,0,-1]]	OU-ISIR center back
sup	r	post	[[0,-1,0],[1,0,0],[0,0,1]]	[[1,0,0],[0,-1,0],[0,0,-1]]	[[0,1,0],[1,0,0],[0,0,-1]]	OU-ISIR center back (smartphone)
post	sup	r	[[1,0,0],[0,1,0],[0,0,1]]	[[0,0,1],[0,1,0],[-1,0,0]]	[[0,0,1],[0,1,0],[-1,0,0]]	OU-ISIR right waist
sup	l	ant	[[0,-1,0],[1,0,0],[0,0,1]]	[[1,0,0],[0,1,0],[0,0,1]]	[[0,-1,0],[1,0,0],[0,0,1]]	DBHuGA - thigh and shin
sup	r	post	[[0,-1,0],[1,0,0],[0,0,1]]	[[1,0,0],[0,-1,0],[0,0,-1]]	[[0,1,0],[1,0,0],[0,0,-1]]	DBHuGA - - center back
sup	post	l	[[0,-1,0],[1,0,0],[0,0,1]]	[[1,0,0],[0,0,1],[0,-1,0]]	[[0,0,-1],[1,0,0],[0,-1,0]]	DBHuGA - - left wrist
ant	inf	r	[[1,0,0],[0,1,0],[0,0,1]]	[[0,0,1],[0,-1,0],[1,0,0]]	[[0,0,1],[0,-1,0],[1,0,0]]	MAREA - right ankle, right wrist
post	inf	l	[[1,0,0],[0,1,0],[0,0,1]]	[[0,0,-1],[0,-1,0],[-1,0,0]]	[[0,0,-1],[0,-1,0],[-1,0,0]]	MAREA - left ankle
r	post	inf	[[1,0,0],[0,0,1],[0,1,0]]	[[1,0,0],[0,-1,0],[0,0,-1]]	[[1,0,0],[0,0,-1],[0,-1,0]]	Vajdi - left front pocket
post	l	inf	[[1,0,0],[0,0,1],[0,1,0]]	[[0,-1,0],[-1,0,0],[0,0,-1]]	[[0,-1,0],[0,0,-1],[-1,0,0]]	Vajdi - right waist